We create chemistry

Safety data sheet

1. Identification

Product identifier

Loxanol® MI 6311

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: additives for inks, varnishes or coatings, stabilizer, Paint additive
Recommended use: Chemical, for industrial use only

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Business Unit Resins and Additives

Telephone: +49 621 60-0
E-mail address: ed-psr@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

No need for classification according to GHS criteria for this product.

Label elements

Globally Harmonized System (GHS)

The product does not require a hazard warning label in accordance with GHS criteria.

Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

imidazole derivative
in water

Hazardous ingredients (GHS)
According to UN GHS criteria

No particular hazards known.

4. First-Aid Measures

Description of first aid measures

Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water.

Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far

Hazards: (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Symptomatic treatment (decontamination, vital functions).

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
water spray, foam, dry powder, carbon dioxide

Special hazards arising from the substance or mixture

nitrogen oxides, carbon oxides
The substances/groups of substances mentioned can be released in case of fire. Under certain conditions in case of fire other hazardous combustion products may be generated.

Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Environmental precautions

Discharge into the environment must be avoided.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Pick up with suitable appliance and dispose of.
Incinerate or take to a special waste disposal site in accordance with local authority regulations.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Keep away from sources of ignition - No smoking.

Conditions for safe storage, including any incompatibilities

Segregate from acids and acid forming substances.
Further information on storage conditions: Containers should be stored tightly sealed in a dry place.

Storage stability:
Storage temperature: <= 40 °C
Storage duration: 12 Months
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

111-41-1: 2-(2-Aminoethylamino)ethanol

Exposure controls

Personal protective equipment

Respiratory protection:
Respiratory protection required in case of exceptional circumstances (e.g.: accidental release, exceeding the occupational exposure limit) Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
butyl rubber (butyl) - 0.7 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
In order to satisfy general industrial hygiene rules safety glasses with side-shields (e.g. EN 166) are recommended.

Body protection:

No body protection required if used for intended purpose and satisfying generally accepted industrial hygiene rules.

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is recommended. Store work clothing separately. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift.

9. Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	colourless to yellowish	
Odour:	amine-like	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Melting point:	< -30 °C	
decomposition point:	> 150 °C (996 hPa) Information based on the main component/s., Product turns dark. The substance / product decomposes.	(OECD Guideline 103)
Lower explosion limit:	For liquids not relevant for classification and labelling.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Flash point:	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2719, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 13736, closed cup)
	No flash point - Measurement made up to pilot light extinguishes.	(ISO 2592, open cup)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Auto-ignition temperature: 375 °C (DIN 51794)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Thermal decomposition: 380 °C, 140 kJ/kg (DSC (OECD 113))

SADT:	Study scientifically not justified.
pH value:	9 - 11 (100 g/l, 20 °C) Information based on the main component/s.

Viscosity, dynamic:	27,4 mPa.s (25 °C)	(ASTM D 7042)
Solubility in water:	Information based on the main component/s. 982 g/l (20 °C, 1.013 hPa, pH 10,4)	(OECD Guideline 105)
Partitioning coefficient n-octanol/water (log Kow):	-0,7 (20 °C; pH value: 10) Information based on the main component/s.	(OECD Guideline 107)
Vapour pressure:	0,000013 hPa (25 °C) Information based on the main component/s.	(OECD Guideline 104)
Density:	1,311 g/cm3 (20 °C, 1.013 hPa) Information based on the main component/s.	(OECD Guideline 109)
Relative vapour density (air):	Water content greater than 10%.	

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Corrosion to metals

Corrosive effects to metal are not anticipated.

Other safety characteristics

Miscibility with water:
completely (e.g. >=90%)

Hygroscopy: Non-hygroscopic

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one

Adsorption/water - soil: *log KOC:* -0,096 *(calculated)*

SAPT-Temperature:
Study scientifically not justified.

10. Stability and Reactivity

Reactivity

Corrosion to metals: Corrosive effects to metal are not anticipated.

Possibility of hazardous reactions

No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid

Avoid extreme heat.

Incompatible materials

Substances to avoid:
strong bases, strong acids

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

(by inhalation):No data available.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one

Experimental/calculated data:
LD50 rat (oral): > 2.000 mg/kg (similar to OECD guideline 401)
Limit concentration test only (LIMIT test). No mortality was observed.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
LD50 rat (dermal): > 5.000 mg/kg (OECD Guideline 402)
Limit concentration test only (LIMIT test). No mortality was observed.

Irritation

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of irritating effects:
Not irritating to the skin. Not irritating to the eyes.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant (OECD Guideline 404)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Serious eye damage/irritation rabbit: non-irritant (OECD Guideline 405)

Respiratory/Skin sensitization

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Experimental/calculated data:
Guinea pig maximization test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of mutagenicity:
The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of carcinogenicity:
No data was available concerning carcinogenic activity.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on available data, the classification criteria are not met.

Developmental toxicity

Assessment of teratogenicity:
Based on available data, the classification criteria are not met.

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
Based on available data, the classification criteria are not met.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of repeated dose toxicity:
Repeated oral uptake of the substance did not cause substance-related effects.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statements on toxicology have been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Toxicity to fish:
LC50 (96 h) 10.250 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 84/449/EWG, C.1, static)
The statement of the toxic effect relates to the analytically determined concentration.

LC50 (96 h) > 1.004 mg/l, Oncorhynchus mykiss (OECD Guideline 203, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Aquatic invertebrates:
EC50 (48 h) > 100 mg/l, Daphnia magna (Directive 79/831/EEC, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Aquatic plants:
EC50 (72 h) > 1.059 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
The statement of the toxic effect relates to the analytically determined concentration.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Microorganisms/Effect on activated sludge:
EC20 (30 min) > 1.000 mg/l, activated sludge, domestic (DIN EN ISO 8192)
Nominal concentration.

Persistence and degradability

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria). Inherently biodegradable. The product is biodegradable after extended adaptation.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Elimination information:
< 10 % DOC reduction (28 d) (DIN EN ISO 7827) (municipal sewage treatment plant effluent)

> 90 % DOC reduction (28 d) (OECD Guideline 302 B) (activated sludge, domestic)

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment of stability in water:
In contact with water the substance will hydrolyse slowly.

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Information on Stability in Water (Hydrolysis):
$t_{1/2}$ > 365 d (pH 7, 25 °C, pH value 7), (calculated)

Bioaccumulative potential

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Information on: 1-(2-Hydroxyethyl)imidazolidin-2-one
Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The local regulations on waste-water treatment must be followed.

13. Disposal Considerations

Waste treatment methods

Incinerate in suitable incineration plant, observing local authority regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Land transport

ADR

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

RID

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Inland waterway transport
ADN

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

Not applicable

16. Other Information

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

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